Week 2: ISA-95, Production Philosophies, and Database Foundations

Deep Dive into Manufacturing Control Systems

ISE 573 – Manufacturing Execution Systems

University of Rhode Island

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Week Overview

ISA-95 Standards

Explore ISA-95 and its role in defining manufacturing system layers

Production Philosophies

Deep dive into production philosophies: Push, Pull, TOC, JIT, CONWIP

Mathematical Foundations

Understand the mathematical foundations of each approach

Database Basics

Learn relational database basics with SQLite

MES Schema

Build a comprehensive MES schema

Practical Implementation

Connect theory to practical implementation

ISA-95 Overview

What is ISA-95?

International standard for the integration of enterprise and control systems

- **Developed by ISA (International Society of Automation)**
- **Also known as ANSI/ISA-95 or IEC 62264**
- **Defines five levels of manufacturing decision-making**
- **Provides consistent terminology and data models**
- **Enables seamless integration between business and manufacturing**

Why ISA-95 Matters

Business Benefits:

- Standardized communication
- Reduced integration costs
- Vendor interoperability
- Clear responsibility boundaries

Technical Benefits:

- Common data models
- Defined interfaces
- Scalable architecture
- Best practices codified

ISA-95 Levels - Detailed

Level	Name	Description
4	Business Planning	ERP, financial planning, long-term strategy (months to years)
3	Manufacturing Operations	MES, scheduling, dispatching, tracking (hours to days)
2	Supervisory Control	SCADA, batch control, continuous control (minutes to hours)
1	Control Systems	PLCs, DCS, real-time control (seconds to minutes)
0	Physical Process	Sensors, actuators, field devices (milliseconds)

Level 3 (MES) - The Integration Hub

Core Functions of MES

Acts as the critical bridge between business planning and shop floor execution

Key Responsibilities:

- Resource allocation and status tracking
- Detailed production scheduling
- Dispatching production orders
- Data collection and acquisition
- Quality management
- Process management
- Maintenance management
- Performance analysis

ISA-95 Object Models

Resources:

- Equipment
- Personnel
- Materials
- Physical assets

Operations:

- Production operations
- Maintenance activities
- Quality tests
- Inventory movements

Information Objects:

Definitions of data exchanged between levels

Discussion: ISA-95 in Your Organization

Discussion Questions

1. Where does your organization fit in the ISA-95 model?
2. What systems operate at each level?
3. Where are the integration gaps?
4. How well does Level 3 (MES) communicate with Level 4 (ERP)?

Take 3 minutes to discuss with a neighbor

Why Study Production Philosophies?

- Define how and when production is triggered
- Influence MES data design and scheduling logic
- Determine inventory policies and WIP levels
- Impact system responsiveness and efficiency
- Guide investment in automation and control systems

Key Insight

The production philosophy you choose fundamentally shapes your MES requirements

Production Control Spectrum

Most real systems use a hybrid approach

Push Systems - Fundamentals

Core Principle

Production is driven by forecasted demand and planned schedules

Characteristics:

- Orders released based on master production schedule (MPS)
- Work pushed through production regardless of downstream capacity
- Material Requirements Planning (MRP) drives component needs
- Inventory buffers between stages

Push System Flow

MRP Logic - The Push Engine

Material Requirements Planning (MRP)

Calculates what, how much, and when to produce based on:

01

Master Schedule: What finished goods are needed

02

Bill of Materials (BOM): What components are required

03

Inventory Records: What's currently on hand

04

Lead Times: How long each operation takes

Output: Planned order releases for all levels of assembly

Push System Database Schema

```
CREATE TABLE MasterSchedule ( schedule_id INTEGER PRIMARY KEY, product_id INTEGER, quantity INTEGER, planned_start DATE, planned_finish DATE);CREATE TABLE PlannedOrders ( order_id INTEGER PRIMARY KEY, part_id INTEGER, quantity INTEGER, release_date DATE, due_date DATE, status TEXT DEFAULT 'Planned');
```

Push Systems - Advantages

When Push Works Well:

- Stable, predictable demand
- Long production lead times
- Complex product structures
- High setup costs
- Make-to-stock environments

Benefits:

- Advance planning visibility
- Material ordering lead time
- Capacity planning possible
- Economies of scale in batching
- Good for seasonal products

Push Systems - Disadvantages

The Forecast Problem

> *All forecasts are wrong; some are useful.* – George Box

Challenges:

- High Work-in-Process (WIP) inventory
- Risk of obsolescence when demand changes
- Long lead times mask problems
- Overproduction of slow-moving items
- Difficulty responding to urgent orders
- Hidden capacity constraints

Push System Metrics

Metric	Formula	Target
Forecast Accuracy	Actual / Forecast × 100%	> 85%
Schedule Adherence	On Time / Total Orders × 100%	> 95%
WIP Turns	Avg WIP Inventory Value	Maximize
Inventory Days	Daily Usage Value	Minimize

Pull Systems - Fundamentals

Core Principle

Production is triggered by actual consumption or downstream demand

Characteristics:

- Nothing is produced until there's a signal (pull signal)
- Downstream processes "pull" from upstream
- Inventory is minimized and controlled
- Flow is balanced across the production line

Key Difference from Push

Pull reacts to actual demand; Push anticipates demand

Pull System Flow

Kanban - The Classic Pull Mechanism

What is Kanban?

Visual signal (card, bin, electronic message) that triggers production or material movement

Types of Kanban:

01

Production Kanban: Authorization to produce

02

Withdrawal Kanban: Authorization to move material

03

Supplier Kanban: Signal to external suppliers

Number of Kanbans = WIP Limit

Kanban Calculation

Kanban Formula

$$K = \frac{D \times L \times (1 + S)}{C}$$

Where:

- K = Number of kanbans
- D = Demand rate (units per time period)
- L = Lead time (replenishment time)
- S = Safety factor (e.g., 0.1 for 10% buffer)
- C = Container capacity (units per kanban)

Example: *D=100/{\rm day}, L=0.5{\rm day}, S=0.1, C=10*

K=\frac{100\times0.5\times1.1}{10}=5.5\approx6 kanbans

Pull System Database Schema

```
CREATE TABLE KanbanCards ( kanban_id INTEGER PRIMARY KEY, part_id INTEGER, from_location TEXT, to_location TEXT,
quantity INTEGER, status TEXT CHECK(status IN ('Available','InUse','InTransit')));CREATE TABLE PullSignals (
signal_id INTEGER PRIMARY KEY, workstation_id INTEGER, part_id INTEGER, timestamp DATETIME DEFAULT CURRENT_TIMESTAMP,
quantity_needed INTEGER, responded BOOLEAN DEFAULT 0);
```

Pull Systems - Advantages

Benefits:

- Minimal WIP inventory
- Quick response to demand changes
- Problems become immediately visible
- Reduced storage space requirements
- Lower risk of obsolescence
- Simplified scheduling
- Better flow synchronization

Toyota's Success

Pull systems enabled Toyota to maintain inventory turns 3-4x higher than competitors

Pull Systems - Disadvantages

Challenges:

- Requires stable, reliable processes
- Vulnerable to supply chain disruptions
- Limited flexibility for demand surges
- Requires continuous improvement culture
- Less effective with high product variety
- Needs reliable equipment and quality
- Difficult to implement with long lead times

Critical Requirement

Pull systems require process capability and reliability

Pull System Metrics

Metric	Formula	Target
Kanban Turns	Consumption / Avg Kanban Inventory	Maximize
Stockout Rate	Stockout Events / Pull Signals	< 2%
WIP Level	Count of active kanbans	Fixed limit
Replenishment Time	Time from signal to availability	Minimize

Just-In-Time (JIT) Philosophy

Definition

Produce and deliver exactly what is needed, when it is needed, in the exact quantity needed

Origins:

- Developed by Toyota (Taiichi Ohno) in 1950s-1970s
- Part of Toyota Production System (TPS)
- Foundation for Lean Manufacturing

Note: JIT is broader than just pull systems – it's a complete manufacturing philosophy

Seven Wastes (Muda) in JIT

- Overproduction
- Waiting
- Transportation
- Over-processing
- Inventory
- Motion
- Defects

JIT Goal

Eliminate all seven wastes to achieve perfect flow

JIT Pillars

Three Fundamental Pillars:

1

Continuous Flow

- Eliminate batch processing
- Reduce lot sizes to one (ideally)
- Balance workstation capacities

2

Takt Time

- Match production rate to customer demand rate
- Takt Time = Available Time / Customer Demand

3

Pull Production

- Use kanban or other pull signals
- Produce only what's consumed

Takt Time Calculation

Takt Time Formula

$$TaktTime = \frac{AvailableProductionTimeperPeriod}{CustomerDemandperPeriod}$$

Example:

- Available time: 480 minutes/day (8 hours)
- Customer demand: 240 units/day
- Takt time = 480 / 240 = 2 minutes/unit

Implication: Every process must complete in ≤ 2 minutes to meet demand

JIT Requirements for Success

Process Requirements:

- High equipment reliability
- Quick changeovers (SMED)
- Zero-defect quality
- Predictable process times
- Minimal variability

Organizational Requirements:

- Cross-trained workers
- Empowered teams
- Continuous improvement (Kaizen)
- Supplier partnerships
- Management commitment

JIT Tracking in Database

```
CREATE TABLE TaktTime ( workstation_id INTEGER PRIMARY KEY, target_takt_time REAL, -- minutes actual_cycle_time REAL, utilization REAL, timestamp DATETIME DEFAULT CURRENT_TIMESTAMP);CREATE TABLE AndonEvents ( event_id INTEGER PRIMARY KEY, workstation_id INTEGER, event_type TEXT, -- Quality, Machine, Material start_time DATETIME, resolution_time DATETIME, downtime_minutes REAL);
```

JIT Metrics

Metric	Formula	Target
Takt Time Achievement	Actual Cycle Time / Takt Time	≤ 1.0
First Pass Yield	Good Units / Total Units × 100%	> 99%
Changeover Time	Minutes from last good to first good	< 10 min
Equipment OEE	Availability × Performance × Quality	> 85%
Lead Time	Days from order to delivery	Minimize

Theory of Constraints - Introduction

Core Premise

Every system has at least one constraint that limits throughput

Developed by: Dr. Eliyahu Goldratt (1984, "The Goal")

Key Insight:

- Improving a non-constraint doesn't improve system throughput
- Focus improvement efforts on the constraint (bottleneck)
- Subordinate everything else to the constraint

The Five Focusing Steps

IDENTIFY the system constraint

Find the bottleneck (lowest capacity resource)

EXPLOIT the constraint

Maximize utilization without additional investment

SUBORDINATE everything else

Adjust other resources to support the constraint

ELEVATE the constraint

Add capacity if needed

REPEAT

Find the next constraint

Drum-Buffer-Rope (DBR) Scheduling

TOC Scheduling Method

Synchronize production flow around the constraint

Components:

- **Drum:** The constraint sets the pace (beat)
- **Buffer:** Time or inventory buffer before constraint
- **Rope:** Signal to control material release

Identifying Constraints in MES

Data-Driven Constraint Identification

Use MES data to find bottlenecks automatically

Indicators of Constraints:

- Highest utilization rate
- Largest queue (WIP accumulation)
- Longest average wait times
- Most frequent schedule delays
- Lowest throughput per hour

TOC Constraint Detection Query

```
-- Find potential constraint by queue sizeSELECT w.workstation_name, COUNT(*) AS queue_size, AVG(o.wait_time_minutes) AS avg_wait, SUM(CASE WHEN o.status='Processing' THEN 1 ELSE 0 END) AS active_jobsFROM Operations oJOIN Workstations w ON o.workstation_id = w.workstation_idWHERE o.status IN ('Waiting','Processing')GROUP BY w.workstation_idORDER BY queue_size DESC, avg_wait DESCLIMIT 1;
```

Buffer Management in TOC

Three Types of Buffers

Protect throughput from variability

1

Constraint Buffer

- Ensures constraint never starves
- Typically 3-5 days of work

2

Shipping Buffer

- Protects on-time delivery

3

Assembly Buffer

- Ensures all components arrive together

TOC Metrics

Metric	Description	Target
Throughput (T)	Rate money is generated through sales	Maximize
Inventory (I)	Money invested in items to sell	Minimize
Operating Expense (OE)	Money spent to convert I to T	Minimize
Financial Metrics:		
Net Profit	T − OE	Maximize
ROI	(T − OE)/I	Maximize